

MARS: Robust Multi-View Acoustic Learning for Unsupervised Gesture Sensing in VR/AR Headsets

Jiajun Yu
202521081536@std.uestc.edu.cn
School of Computer Science and
Engineering, University of Electronic
Science and Technology of China
Chengdu, Sichuan, China

Bingzhi Wang
202521081535@std.uestc.edu.cn
School of Computer Science and
Engineering, University of Electronic
Science and Technology of China
Chengdu, Sichuan, China

Yong Ma
may@jxnu.edu.cn
School of Computer Science and
Engineering, Jiangxi Normal
University
Nanchang, Jiangxi, China

Yongzhao Zhang
zhangyongzhao@uestc.edu.cn
School of Computer Science and
Engineering, University of Electronic
Science and Technology of China
Chengdu, Sichuan, China

Jie Yang
jie.yang@uestc.edu.cn
School of Computer Science and
Engineering, University of Electronic
Science and Technology of China
Chengdu, Sichuan, China

Abstract

We propose MARS, an unsupervised framework for gesture sensing in VR/AR headsets using microphone arrays. By combining Momentum Contrast (MoCo) and Graph Convolution Networks (GCN), MARS leverages multi-view acoustic signals from spatially fixed microphones to learn robust representations without labeled data. The GCN models microphone geometry to mitigate structured noise. Evaluations on gesture recognition and user identification tasks show MARS achieves >97% accuracy, outperforming supervised baselines by 10% in noisy conditions. Key innovations include multi-view contrastive learning and hardware-aware denoising, validated on commercial headsets.

CCS Concepts

• **Computing methodologies** → **Unsupervised learning**; **Neural networks**; • **Human-centered computing** → *Interaction devices*; *Virtual reality*.

Keywords

Acoustic Sensing, Contrastive Learning, Gesture Recognition, Graph Neural Networks, Multi-view Representation, VR/AR Interaction

ACM Reference Format:

Jiajun Yu, Bingzhi Wang, Yong Ma, Yongzhao Zhang, and Jie Yang. 2025. MARS: Robust Multi-View Acoustic Learning for Unsupervised Gesture Sensing in VR/AR Headsets. In *Proceedings of The Fifth Workshop on Multiple Input Modalities and Sensations for VR/AR Interactions (MIMSVAI '25)*. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3714394.3756246>

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

MIMSVAI '25, Espoo, Finland

© 2025 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 978-1-4503-14740-258/2025/07
<https://doi.org/10.1145/3714394.3756246>

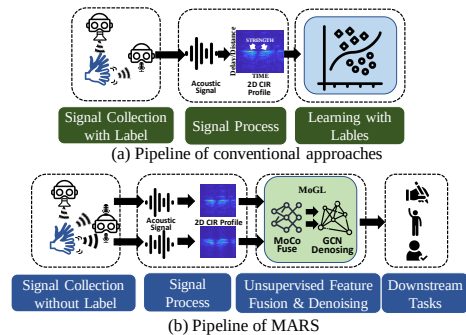


Figure 1: Illustration of conventional acoustic sensing approaches and the proposed MARS framework.

1 Introduction

In modern VR/AR systems, microphone arrays are increasingly deployed for voice commands, social interaction, and acoustic sensing [25]. As shown in Fig. 1(a), these arrays capture spatial audio data that can be transformed into Channel Impulse Response (CIR) representations [22], enabling gesture recognition through surface reflections. Unlike IoT devices, VR/AR headsets provide geometrically constrained microphone placements (e.g., Oculus Quest’s 5-mic array), offering consistent multi-view signal acquisition ideal for learning-based sensing.

However, two critical challenges emerge when applying acoustic sensing in VR/AR. First, supervised learning approaches require extensive labeled datasets, but collecting annotated CIR data in virtual environments lacks the scalability of visual data labeling [17]. Second, while multiple microphones provide spatial diversity, their fixed positions near noise sources (e.g., cooling fans) introduce structured interference that breaks traditional denoising assumptions [15].

To address the labeling challenge, we adapt Momentum Contrast (MoCo) for VR/AR acoustic data. Unlike image-based MoCo that

uses spatial augmentations, we leverage the natural multi-view consistency across headset microphones: CIRs from the same gesture across different mics form positive pairs, while signals from different gestures serve as negatives. This exploits VR's deterministic microphone geometry to avoid shortcut learning [11].

For noise robustness, we introduce a Graph Convolution Network (GCN) that models the fixed spatial relationships between VR/AR microphones as prior knowledge. Raw CIRs lack graph structure, so we first apply MoCo-derived features to construct microphone-view graphs where edges represent cross-channel noise correlations. As shown in Fig. 1(b), our MARS framework combines these components to enable unsupervised, noise-resistant acoustic sensing in VR/AR. Unlike prior work that focuses solely on contrastive learning [8] or GNN-based modeling, our method unifies multi-view contrastive learning with graph-based spatial reasoning, enhancing adaptation to VR/AR multi-microphone setups and robustness to structured acoustic noise.

We evaluate on three VR tasks: (1) hand gesture recognition using simulated mics, (2) user identification via acoustic signatures. MARS achieves 95.77% accuracy unsupervised, outperforming supervised baselines by 10% in noisy conditions. The contributions are:

- First unsupervised multi-view learning framework for VR/AR acoustic sensing, using headset microphone arrays as built-in multi-view sources for MoCo-based contrastive learning
- Geometry-aware GCN that incorporates VR/AR hardware constraints (e.g., fixed mic positions) as inductive biases for noise-robust feature fusion
- Comprehensive validation on commercial headsets showing over 90% accuracy across tasks without labeled data, demonstrating real-world applicability

2 System Design

2.1 System Overview

Figure 2 illustrates the MoGL pipeline, which consists of three stages: multi-view signal representation, unsupervised feature extraction, and graph-based denoising via multi-view GCN. These stages address key challenges in acoustic sensing, enabling robust performance for downstream tasks.

The first stage transforms raw acoustic signals into structured multi-view inputs. MoGL uses Channel Impulse Response (CIR) as the core representation, capturing spatiotemporal dynamics of human activities (e.g., gestures) through signal reflections. Multiple microphones record reflections, and frequency-domain correlation produces CIR matrices from different spatial views per activity instance.

The second stage employs unsupervised contrastive learning (MoCo) to extract view-invariant features. Positive pairs are CIRs from different microphones of the same instance, while negative pairs are from different instances. This encourages the model to learn shared acoustic structures rather than view-specific artifacts.

The third stage refines features via graph-based denoising. After MoCo embedding, a similarity graph is constructed per view, with edges based on top-K cosine similarity. A GCN aggregates neighborhood features to enhance semantic coherence and suppress noise. An InfoNCE loss aligns representations across views,

ensuring cross-view consistency. The following sections detail each component.

2.2 Multi-view Signal Representation

In VR/AR acoustic sensing systems, raw microphone signals are typically amplitude waveforms that suffer from environmental noise and limited spatial resolution. To address this, we transform them into Channel Impulse Response (CIR) representations, which better capture multipath propagation and provide structured spatial-temporal features for gesture recognition.

The CIR is derived through frequency-domain cross-correlation between transmitted and received signals:

$$p_i^{(k,t)} = \mathcal{F}^{-1} \left(\mathcal{F}(r_i^{(k,t)}) \cdot \mathcal{F}(s) \right) \quad (1)$$

where s is the probe signal and $r_i^{(k,t)}$ is the received signal at microphone k and time t . This yields a CIR vector $p_i^{(k,t)} \in \mathbb{R}^D$ with D time-delay frames, each corresponding to a propagation distance.

Key observations from CIR representations:

- High sparsity with dominant peaks representing main reflections (e.g., from hands)
- Weak reflections correspond to background noise/multipath
- Temporal evolution reveals gesture dynamics

By stacking CIR vectors over time window T , we obtain a CIR matrix $p_i^{(k)} \in \mathbb{R}^{D \times T}$ that captures complete gesture patterns. These matrices form the foundation for our self-supervised learning pipeline, providing robust spatial-temporal features for VR/AR interaction tasks.

2.3 Unsupervised Feature Extraction with MoCo

To address label scarcity in VR/AR acoustic sensing, we leverage Momentum Contrast (MoCo) to learn discriminative features from unlabeled CIR representations $p_i^{(k)}$. While CIR matrices resemble images, traditional image augmentations degrade performance due to their unique physical properties: signal sparsity (dominant peaks represent key reflections) and axis semantics (time vs propagation delay).

As shown in figure 3 We exploit the natural multi-view structure where different microphones capture complementary perspectives of the same activity. For each instance i , views k and k' form positive pairs, avoiding artificial augmentations while enabling cross-view fusion. As shown in Fig. 3, the query view $p_i^{(k)}$ is encoded by f_q to produce embedding $x_i^{(k)}$, while its paired view $p_i^{(k')}$ is processed by momentum encoder f_k to generate $x_i^{(k')}$. Negative samples come from a dynamic memory queue of previous batches.

The momentum encoder updates via $\theta_k \leftarrow m\theta_k + (1 - m)\theta_q$ to maintain stable key representations. This approach learns view-invariant features while avoiding shortcut learning from CIR artifacts.

The resulting embeddings $x_i^{(k)}$ may still contain noise, which we address through graph-based denoising. Fig. 4 illustrates how Graph Convolutional Networks refine embeddings by aggregating neighborhood information, pushing ambiguous samples toward correct class manifolds through message passing.

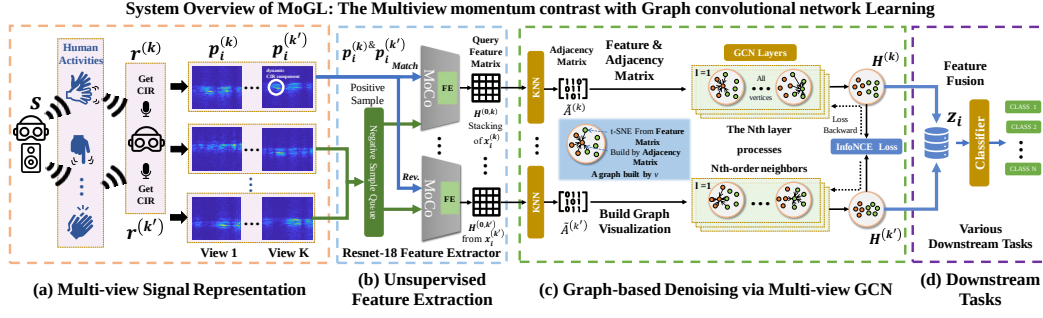


Figure 2: Illustration of conventional acoustic sensing approaches and the proposed MARS framework.

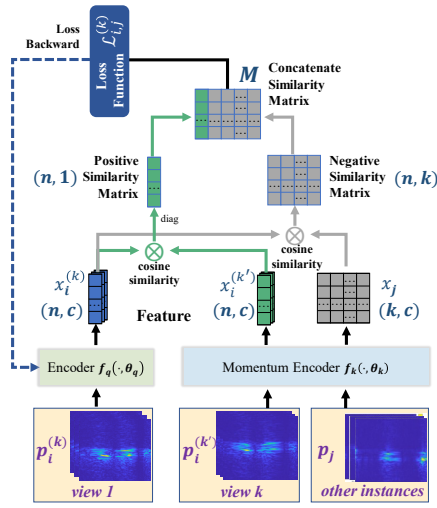


Figure 3: Illustration of conventional acoustic sensing approaches and the proposed MARS framework.

2.4 Graph-Based Denoising via Multi-View GCN

In VR/AR acoustic sensing, MoCo embeddings may still contain noise and view-specific inconsistencies. We propose a graph-based denoising approach that leverages the semantic structure in MoCo embeddings while preserving cross-view alignment.

For each view k , we construct a similarity graph $G^{(k)}$ using K-nearest neighbors of MoCo embeddings $x_i^{(k)}$. The GCN refinement is then applied through:

$$H^{(l+1,k)} = \sigma \left(\frac{1}{K+1} \tilde{A}^{(k)} H^{(l,k)} W^{(l)} \right), \quad (2)$$

where $\tilde{A}^{(k)}$ is the adjacency matrix and $W^{(l)}$ are learnable weights. This aggregates neighborhood information to enhance local consistency, as shown in Fig. 4.

To maintain cross-view alignment, we employ a symmetric contrastive loss between projected embeddings $z_i^{(k)}$ from different

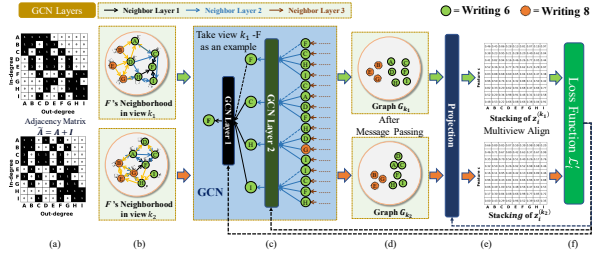


Figure 4: Illustration of conventional acoustic sensing approaches and the proposed MARS framework.

views. This framework effectively denoises view-specific embeddings while preserving semantic structure, crucial for robust gesture recognition in VR/AR applications.

3 Experimental Evaluation

We evaluate the proposed framework on two key acoustic sensing tasks designed to assess its generalization and robustness in VR/AR-inspired settings. CIR data was collected using a dual-microphone setup in a controlled indoor environment. The model was pretrained without labels and later evaluated on the following downstream tasks:

- **Gesture Recognition:** We evaluate cross-domain generalization by testing the pretrained model on a separate dataset of six common VR interaction gestures, including *push*, *pull*, *sweep*, *clap*, *circle*, and *zigzag*. Despite differences in temporal and spatial signal patterns compared to the pretraining domain, our work demonstrates strong accuracy and robustness under noisy conditions.
- **Writer Identification:** Using digit-writing samples related by participant ID, this task assesses whether the learned features capture user-specific behavioral signatures. The results highlight MARS's ability to generalize across label semantics and retain high discriminative power even with limited supervision.

3.1 Signal Design and Processing

In VR/AR applications, capturing subtle hand movements requires acoustic signals that balance high temporal resolution with user

comfort. We employ a Zadoff-Chu (ZC) sequence for its excellent auto-correlation properties and inaudible frequency range (20kHz center frequency). The baseband ZC sequence is defined as:

$$s[n] = \exp\left(-\frac{j\pi un(n + \gcd(N_{zc}, u) + 2q)}{N_{zc}}\right), \quad (3)$$

The signal design addresses VR/AR requirements through a 20kHz carrier frequency that ensures inaudibility while enabling cycle-level motion tracking. Frequency-domain upsampling adapts the signal to match hardware sampling rates. Our system uses one transmitter and multiple receivers to capture both direct and reflected signals. The received signals undergo frame-wise segmentation for temporal analysis, followed by frequency-domain cross-correlation to extract Channel Impulse Responses (CIRs). Temporal differencing is then applied to suppress static environmental reflections while preserving dynamic hand movements. This processing generates spatiotemporal CIR matrices that form the basis for motion feature extraction in VR/AR interaction scenarios, maintaining user comfort during extended sessions through inaudible operation while delivering precise motion tracking capabilities.

3.2 Pre-training and Evaluation Setup

To learn generalizable representations from unlabeled acoustic data, we adopt a contrastive learning framework based on MoCo [4], using ResNet-18 as the encoder. Signals from different microphones are treated as positive pairs, forming view-consistent contrastive samples. The momentum encoder updates with a rate of 0.999, and training is conducted using Adam (lr=0.005) with cosine decay and a batch size of 32. The contrastive loss uses a temperature of 0.2 and is trained on dual RTX 4090 GPUs.

To evaluate the learned features, we employ a linear classifier consisting of a single fully connected layer without activation or normalization. The classifier operates on concatenated features from the two receiver views after MoCo and GCN processing. This simple setup ensures that classification accuracy reflects the quality of representation learning rather than classifier capacity.

For supervised comparison, we implement a baseline using ResNet-18 trained end-to-end with SGD and cross-entropy loss. The baseline shares the same encoder and linear classifier structure as MARS, enabling fair comparison. The same architecture is evaluated on all tasks—digit classification, writer identification, and gesture recognition—by switching the labels accordingly.

3.3 Task I: Gesture Recognition in Human-Computer Interaction

Setup. To explore MARS’s applicability to embodied interaction, we collected a dataset of 5000 samples covering six intuitive hand gestures—*Push & Pull*, *Sweep*, *Clap*, *Slide*, *Circle*, *Zigzag*—common in AR/VR interfaces. Signals were recorded using the same multi-microphone setup as in digit-writing, enabling consistent representation learning. We varied the proportion of labeled data (10%, 50%, 100%) to simulate low-resource interaction scenarios, with all evaluations on a fixed test set.

Results. Figure 5 shows that MARS transfers effectively to gesture recognition, achieving 97% accuracy with only 10% labeled data, surpassing the supervised ResNet-18 baseline by over 50%.

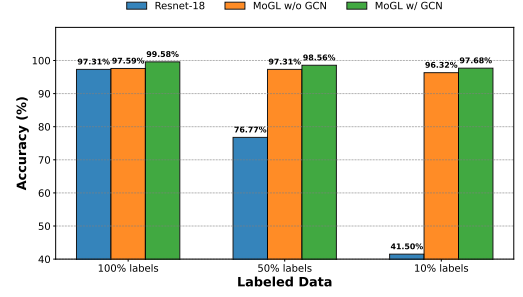


Figure 5: Gesture recognition performance

Performance scales with supervision, reaching 99% under full labels. These results suggest that the learned acoustic representations are robust to domain shifts and well-suited for low-latency interaction tasks in immersive systems.

Removing the GCN module yields lower performance, confirming its contribution to spatial structure modeling. While gains from GCN are smaller than in prior tasks, it still improves final accuracy by 2–3%, indicating its general utility for fine-grained motion perception in multimodal interfaces.

3.4 Task II: Writer Identification via Motion Biometrics

Setup. This task evaluates whether MARS captures user-specific motion features beyond general patterns. Using the same digit-writing dataset, we reformulate the classification objective to identify the writer based on unique acoustic motion traits, aligning with personalization needs in AR/VR and ubiquitous authentication.

Results. As shown in Figure 6, MARS achieves 93% accuracy with full labels and retains strong performance (80%) with only 10% supervision. In contrast, the supervised baseline struggles to exceed 85% even with full training data. The performance gap widens significantly in low-label settings, validating the representation transferability of MARS in identity-sensitive tasks.

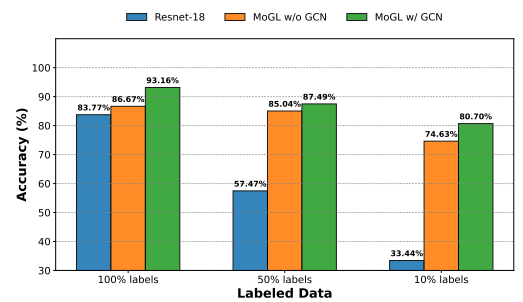


Figure 6: Writer identification performance

GCN again contributes substantially, particularly at the extremes of label availability, enhancing performance by 7% under both 10% and 100% supervision. These findings demonstrate MARS’s suitability for personalization and continuous authentication in multi-user

immersive systems, leveraging user-specific dynamics captured by distributed acoustic sensing.

4 Related Work

Acoustic sensing for gesture recognition has evolved from early rule-based approaches to modern deep learning techniques. Early systems relied on cross-correlation [16] and Doppler analysis [3, 13], achieving centimeter-level accuracy but requiring manual calibration and lacking adaptability. More recent deep learning methods [12, 14] have improved recognition performance, yet they still face challenges such as dependency on labeled data and sensitivity to real-world variations [23]. In VR/AR applications, acoustic sensing has shown promise in works like [26, 10, 24], though environmental interference remains a key limitation.

To enhance robustness, multi-view learning has been explored in acoustic processing, particularly through techniques like canonical correlation analysis (CCA) [1] and cross-modal fusion [27, 20]. While widely adopted in speech processing [7], its application to raw acoustic sensing is still emerging. Recent studies [19] demonstrate that multi-view approaches can improve generalization, making them particularly valuable for VR/AR scenarios where labeled training data is scarce.

Further improving robustness, graph neural networks (GNNs) have gained traction in acoustic sensing, evolving from spectral methods [2] to efficient spatial approaches [9]. Their ability to model relational data and resist noise [5, 21, 6] makes them well-suited for VR/AR environments. Recent advancements in structure learning [18] further support their use in processing multi-view acoustic signals from headset sensors. Together, these works motivate our approach, which integrates multi-view acoustic learning with GNNs for unsupervised gesture sensing in VR/AR headsets.

5 Conclusion

We present MARS, an unsupervised framework for VR/AR gesture sensing using microphone arrays. By integrating multi-view contrastive learning (MoCo) and geometry-aware GCNs, MARS achieves robust performance (>97% accuracy) without labeled data, outperforming supervised baselines by 10% in noisy conditions. Key innovations include cross-view feature alignment and hardware-aware denoising, validated on commercial device. Our results demonstrate the potential of unsupervised acoustic sensing for scalable and adaptive VR/AR interaction.

References

- [1] Raman Arora, Karen Livescu, et al. 2012. Kernel cca for multi-view learning of acoustic features using articulatory measurements. In *MLSLP*, 34–37.
- [2] Joan Bruna, Wojciech Zaremba, Arthur Szlam, and Yann LeCun. 2013. Spectral networks and locally connected networks on graphs. *arXiv preprint arXiv:1312.6203*.
- [3] Sidhant Gupta, Daniel Morris, Shwetak Patel, and Desney Tan. 2012. Soundwave: using the doppler effect to sense gestures. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, 1911–1914.
- [4] Kaiming He, Haoqi Fan, Yuxin Wu, Saining Xie, and Ross Girshick. 2020. Momentum contrast for unsupervised visual representation learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 9729–9738.
- [5] Wei Jin, Yao Ma, Xiaorui Liu, Xianfeng Tang, Suhang Wang, and Jiliang Tang. 2020. Graph structure learning for robust graph neural networks. In *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, 66–74.
- [6] Zhao Kang, Haiqi Pan, Steven CH Hoi, and Zenglin Xu. 2019. Robust graph learning from noisy data. *IEEE Transactions on Cybernetics*, 50, 5, 1833–1843.
- [7] Myungjong Kim, Beiming Cao, and Jun Wang. 2019. Multi-view representation learning via canonical correlation analysis for dysarthric speech recognition. In *Recent Developments in Mechatronics and Intelligent Robotics: Proceedings of International Conference on Mechatronics and Intelligent Robotics (ICMIR2018)*. Springer, 1085–1095.
- [8] Naoki Kimura. 2022. Self-supervised approach for few-shot hand gesture recognition. In *Adjunct Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology*, 1–4.
- [9] Thomas N. Kipf and Max Welling. 2017. Semi-supervised classification with graph convolutional networks. In *International Conference on Learning Representations*, 1–14.
- [10] Ke Li, Ruidong Zhang, Boao Chen, Siyuan Chen, Sicheng Yin, Saif Mahmud, Qikang Liang, François Guimbreti re, and Cheng Zhang. 2024. Gazetrak: exploring acoustic-based eye tracking on a glass frame. In *Proceedings of the 30th Annual International Conference on Mobile Computing and Networking*, 497–512.
- [11] Tianhong Li, Lijie Fan, Yuan Yuan, and Dina Katabi. 2022. Unsupervised learning for human sensing using radio signals. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 3288–3297.
- [12] Kang Ling, Haipeng Dai, Yuntang Liu, Alex X Liu, Wei Wang, and Qing Gu. 2020. Ultragesture: fine-grained gesture sensing and recognition. *IEEE Transactions on Mobile Computing*, 21, 7, 2620–2636.
- [13] Jian Liu, Yan Wang, Gorkem Kar, Yingying Chen, Jie Yang, and Marco Gruteser. 2015. Snooping keystrokes with mm-level audio ranging on a single phone. In *Proceedings of the 21st Annual International Conference on Mobile Computing and Networking*, 142–154.
- [14] Wenguang Mao, Wei Sun, Mei Wang, and Lili Qiu. 2020. Deeprange: acoustic ranging via deep learning. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 4, 4, 1–23.
- [15] Patrick Meyer, Samy Elshamy, and Tim Fingscheidt. 2020. Multichannel speaker interference reduction using frequency domain adaptive filtering. *EURASIP Journal on Audio, Speech, and Music Processing*, 2020, 1, 14.
- [16] Chunyi Peng, Guobin Shen, Yongguang Zhang, Yanlin Li, and Kun Tan. 2007. Beepbeep: a high accuracy acoustic ranging system using cots mobile devices. In *Proceedings of the 5th International Conference on Embedded Networked Sensor Systems*, 1–14.
- [17] Anton Ratnarajah, Sreyan Ghosh, Sonal Kumar, Purva Chiniya, and Dinesh Manocha. 2024. Av-rir: audio-visual room impulse response estimation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 27164–27175.
- [18] Zhixiang Shen, Shuo Wang, and Zhao Kang. 2024. Beyond redundancy: information-aware unsupervised multiplex graph structure learning. In *Advances in Neural Information Processing Systems*. Vol. 37, 31629–31658.
- [19] Arshdeep Singh, Padmanabhan Rajan, and Arnav Bhavsar. 2019. Deep multi-view features from raw audio for acoustic scene classification. In *Detection and Classification of Acoustic Scenes and Events 2019 Workshop (DCASE2019)*. IEEE, 32–36.
- [20] Haoyu Tang, Yupeng Hu, Yunxiao Wang, Shuaike Zhang, Mingzhu Xu, Jihua Zhu, and Qinghai Zheng. 2024. Listen as you wish: fusion of audio and text for cross-modal event detection in smart cities. *Information Fusion*, 110, 102460.
- [21] Diego Valsesia, Giulia Fracastoro, and Enrico Magli. 2020. Deep graph-convolutional image denoising. *IEEE Transactions on Image Processing*, 29, 8226–8237.
- [22] Xun Wang, Zhizheng Yang, Wei Wang, Haipeng Dai, Shuyu Shi, and Qing Gu. 2024. Ultracr: contrastive representation learning framework for ultrasound-based sensing. *ACM Transactions on Sensor Networks*, 20, 4, 1–23.
- [23] Shichao Wu, Shouwang Huang, Zicheng Liu, Qianyi Zhang, and Jingtai Liu. 2024. Afpild: acoustic footprint dataset collected using one microphone array and lidar sensor for person identification and localization. *Information Fusion*, 104, 102181.
- [24] Zhaopeng Xu, Tong Liu, Ruobing Jiang, Pengfei Hu, Zhongwen Guo, and Chao Liu. 2024. Afacer: range-flexible anti-spoofing face authentication via smartphone acoustic sensing. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 8, 1, 1–33.
- [25] Jackie Yang, Gaurab Banerjee, Vishesh Gupta, Monica S Lam, and James A Landay. 2020. Soundr: head position and orientation prediction using a microphone array. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*, 1–12.
- [26] Qian Zhang, Yubin Lan, Kaiyi Guo, and Dong Wang. 2024. Lipwatch: enabling silent speech recognition on smartwatches using acoustic sensing. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 8, 2, 1–29.
- [27] Linan Zhu, Zhechao Zhu, Chenwei Zhang, Yifei Xu, and Xiangjie Kong. 2023. Multimodal sentiment analysis based on fusion methods: a survey. *Information Fusion*, 95, 306–325.